AFA Platform - Geographic Information System

ISKOftware

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Project blog:
https://miriitode.github.io/CS191-OFA-GIS/

CS 191 WFR1
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1. Customer State of Requirements

Project Motivation

The OFA (Office of Field Activities) is an administrative unit of the University of the Philippines Diliman. This unit promotes and manages academic field activities that can happen outside of the University to enrich students' education, expose them to diverse communities, and encourage them to serve their nation. This unit is also mainly responsible for maintaining policies and guidelines for academic field activities, making sure that said academic field activities are conducted within the context of the course or curriculum they originate from, and ensuring the safety of the students and faculty participating in these activities.

According to the OFA, Academic Field Activities (AFAs) are experimental learning experiences and activities conducted in fulfillment of the requirements, expectations, and learning outcomes of a given course or subject. These activities usually involve learning through activities that are uniquely conducted outside of the usual in-class learning, in order to further enrich the student's knowledge of the subject and potentially apply their knowledge in a practical context.

For AFAs to be properly conducted, academic units and their faculties must consult with the OFA to review the activity, provide plans and specific details of the activity at hand, and then must be approved by the OFA. The office has to handle a lot of documentation, which includes approval documents and acknowledgement documents for approved AFAs. The office also has to track these documents, such as when they are in transit from one department to another, or the statuses of the approval process of proposed AFAs.

The current solution of the office is that they work on a centralized spreadsheet that contains all of the documents that the office holds. The spreadsheet tracks the type of AFA, the department or unit it originated from, the approval process or state, the location where the AFA will take place, and any extra information related to the AFA.

The Problem

Applying for and seeking approval for a proposed AFA involves filling out forms and providing the necessary information. There are numerous forms for each of the different types of AFA, and they are separate documents as a result. These different kinds of forms have to be documented and be marked in the spreadsheet, resulting in applicants having to identify the right category to which the AFA belongs before filling up the corresponding proposal form beforehand.

Additionally, the OFA wants to have a Geographical Information System, an interface that shows a map that marks any approved AFAs in a given area and/or time. However, they do not have an existing system yet.

Thus, the OFA wants to have a centralized system for all the documents, as well as a unified portal for units to submit AFA proposals, a way for students to check and apply for partnered institutions and companies, and a way for visitors to see the GIS and the various AFAs that the university has, past and present.

How Current Practices Solve the Problem

In essence, the GIS project is as simple as making use of an interactive map viewer with map pins. There are a lot of libraries for interactive maps that are available for use, for example:

- **Leaflet** is an open-source JavaScript library for interactive maps. (https://leafletjs.com/) It is extremely lightweight and offers a lot of features, such as easy yet powerful customization, flexible content for map markers, and such. It has the added benefit of being friendly for mobile devices, which can benefit this project greatly.

- **MapLibre GL JS** is an open-source TypeScript Library for interactive maps. (https://maplibre.org/maplibre-gl-js/docs/) It renders vector tiles for its maps. Unlike raster tiles, which use pre-rendered images of maps that are stitched together to create the viewable map, vector tiles make use of compressed raw data representing the map, and are rendered on the fly. This allows for much faster loading of the map, since the data for vector tiles are much smaller than raster tiles. It also allows for more control over the design and aesthetics of the map, such as being able to rotate and tilt maps or to be able to see the map in 3D. It does have the downside of it being more processor intensive to render the map compared to raster tiles. (https://www.thunderforest.com/articles/vector-vs-raster-maps/)

- **OpenLayers** is an open-source JavaScript library for interactive maps. (https://openlayers.org/) It is similar to Leaflet, but has many more features that are beneficial for complex GIS applications.

- **Google has its own API for its Maps**. (https://mapsplatform.google.com/lp/maps-apis/) It allows for creating simple, static maps to complex and immersive maps. Since it is an API, it can be easy to implement, but with the downside of having to depend on an external API, with costs added depending on how many API calls are made.

System Requirements

The requirement of this project is to create the GIS section of the website. Specifically:

1. To properly display all AFAs from the current existing database and their information, such as the name of the activity and the location (both specific details and a physical location) on an interactive map. Ideally, each AFA can be represented as a pinned location on an interactive map. When the pin is clicked or tapped, it will display the full

details of the AFA, such as the name of the AFA, the section and department for the AFA, and such.

2. To show the AFAs from a given time period or academic year. The user can use a slider or a date input, and the website will filter out and display AFAs for a specific time period. It also has options for the user to use a preset date range in the form of academic years and semesters for easier filtering. Consequently, the GIS must properly update the display of AFA locations corresponding to the user's input for the date range.

3. Additionally, this part of the website is to be integrated with the entire proposed AFA platform website. The website must be both compatible with desktop and mobile devices and must be simple to use and smooth for a great user experience.

2. Preliminary UI Design

The user interface of the GIS will revolve around a map viewer where both users and administrators can access it with similar functionalities. Since the data shown is based on the data of the AFA database, the GIS will automatically be able to show pinpoint locations of each AFA, with relevant information included.

When the page is first opened, the map will default to the entire map of the Philippines. The default date parameter is set to the current date and current academic year, and so the AFAs that happened from the start of the academic year up to the current date. Figure 1 shows this scenario:

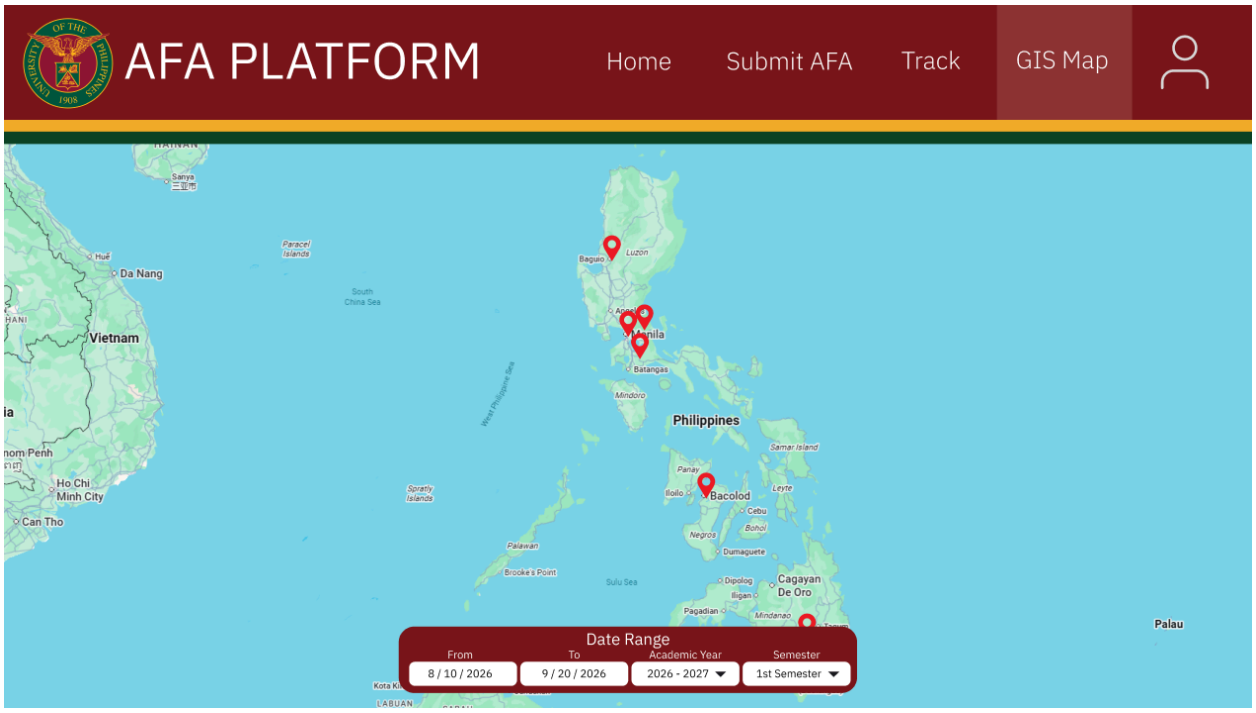


Figure 1: AFA - GIS Page on first visit

The date range and academic / semesters can be freely changed using the tab at the bottom, and what shows up on the GIS will prioritize what was last modified from the range settings.

Hovering on a pin will show basic details of the AFA, such as the name and date. Clicking / tapping on that pin will show a side panel with more relevant information. Figures 2 and 3 show these scenarios:

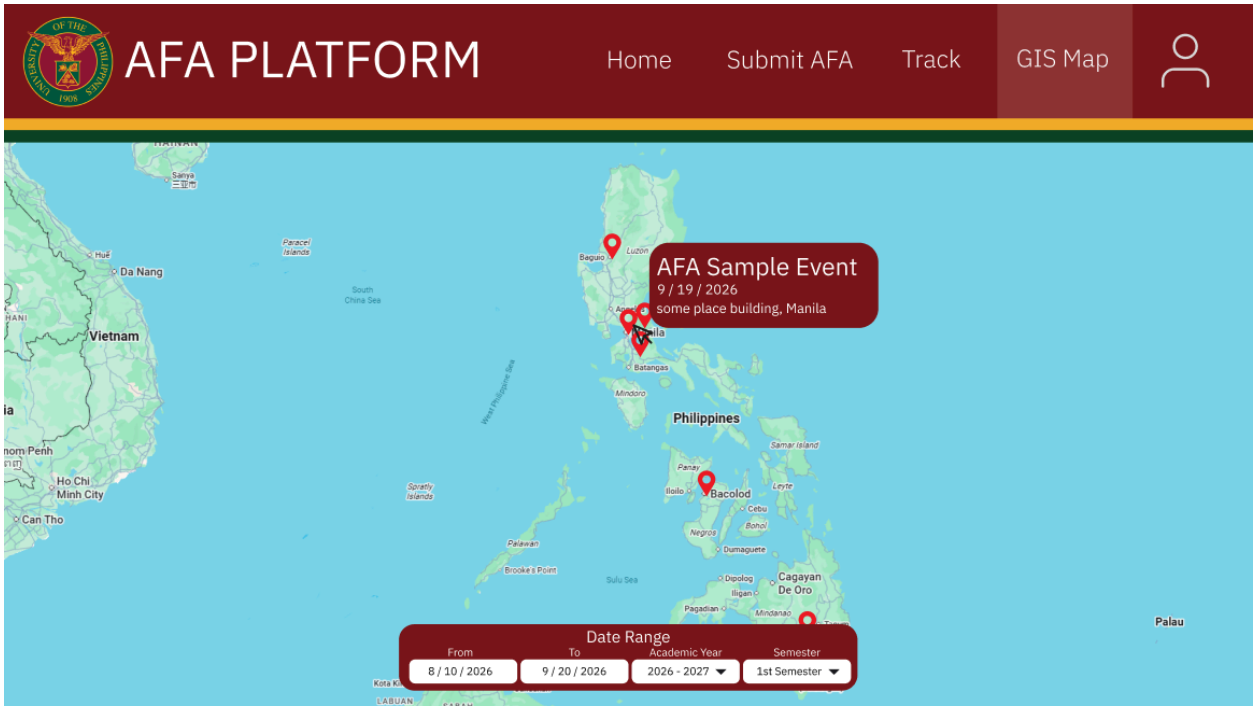


Figure 2: AFA - GIS Page with quick information on pin

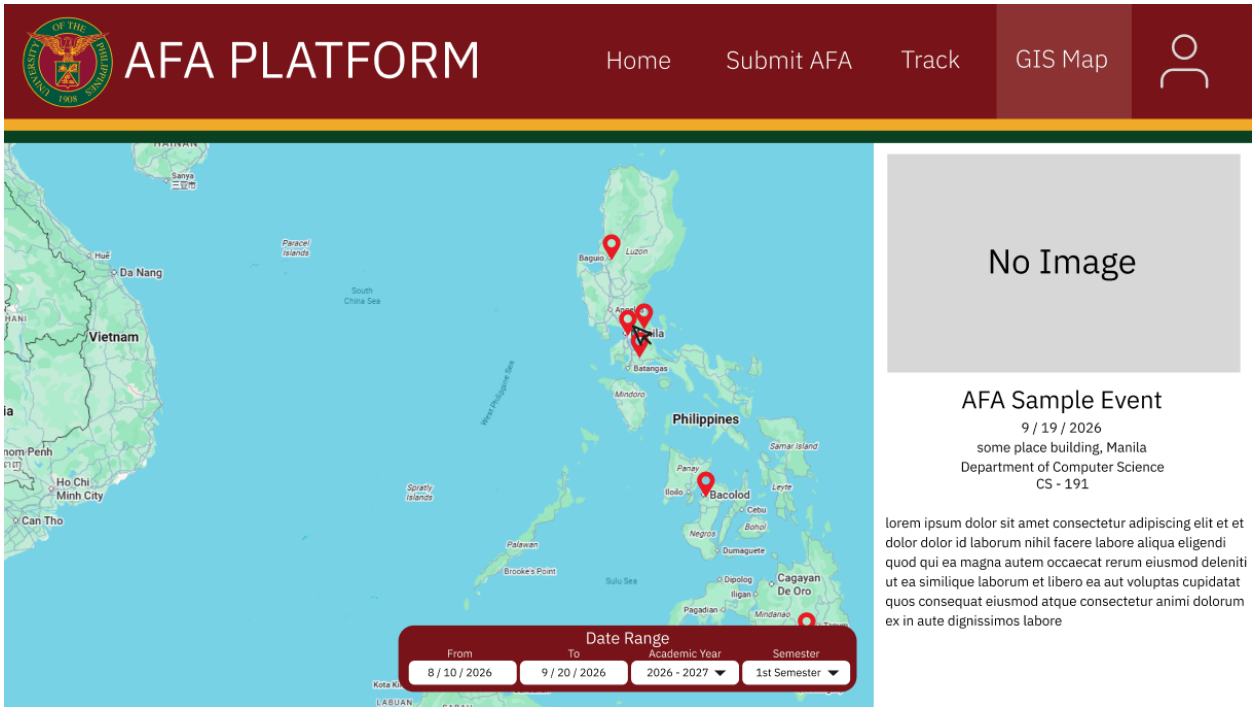


Figure 3: AFA - GIS Page with detailed information

3. Plan of Work

3.1. Estimated Cost and Resources

To establish a reliable project baseline and mitigate scheduling risks, we will be using estimation techniques from Pressman's "Software Engineering: A Practitioner's Approach." For this project, we will be allocating resources to a target budget of 440 total person-hours.

Additionally, because the development of this project will be relying entirely on open-source frameworks and free-tier infrastructure, no direct financial costs will be incurred.

Process-based Estimation

This technique distributes project hours across standard activities of the development cycle, allocating estimated effort across Communication, Planning, Database Design, Construction, and Testing. This ensures that non-coding activities are also properly budgeted alongside actual development.

Activity	**Estimated Effort**	**Person-Hours**	**Timeline Placement**
Communication	15%	66 hours	Weeks 1-16
Planning	15%	66 hours	Weeks 1-3
Database Design	25%	110 hours	Weeks 4-6
Construction	35%	154 hours	Weeks 7-11
Testing	10%	42 hours	Weeks 12-14
Total	**100%**	**440 hours**	**16 Weeks**

Problem-based Estimation

This budgets effort by decomposing the system into its core functional components: Database, Backend API, UI/UX Design & Map Viewer, and Integration & Testing. This will allow us to evaluate the complexity of each feature and verify that our estimates align with the budgeted effort hours.

Component	**Description**	**Person-Hours**
Database	Setting up spatial schemas, handling data pipelines to process records	120 hours

Backend API	Developing server-side logic to process AFA location data	110 hours
UI/UX Design & Map Viewer	Rendering map design and pins, data popups, etc.	120 hours
Integration and Testing	Performing query unit tests, debugging, embedding into platform	90 hours
Total	**Sum of all components**	**440 hours**

Minimum System Resources

- **Disk Storage:** At least 5 GB of free space
- **Memory (RAM):** 4 GB RAM
- **Display:** Color display with a minimum resolution of 1366 x 768
- **Network:** Stable bandwidth of at least 5 Mbps download/upload speed

3.2. Schedules and Milestones

To illustrate the schedule for the development of the project, a Gantt chart (Figure 4) will be utilized showing when certain milestones will be done. Week 1 of the timeline starts on October 5.

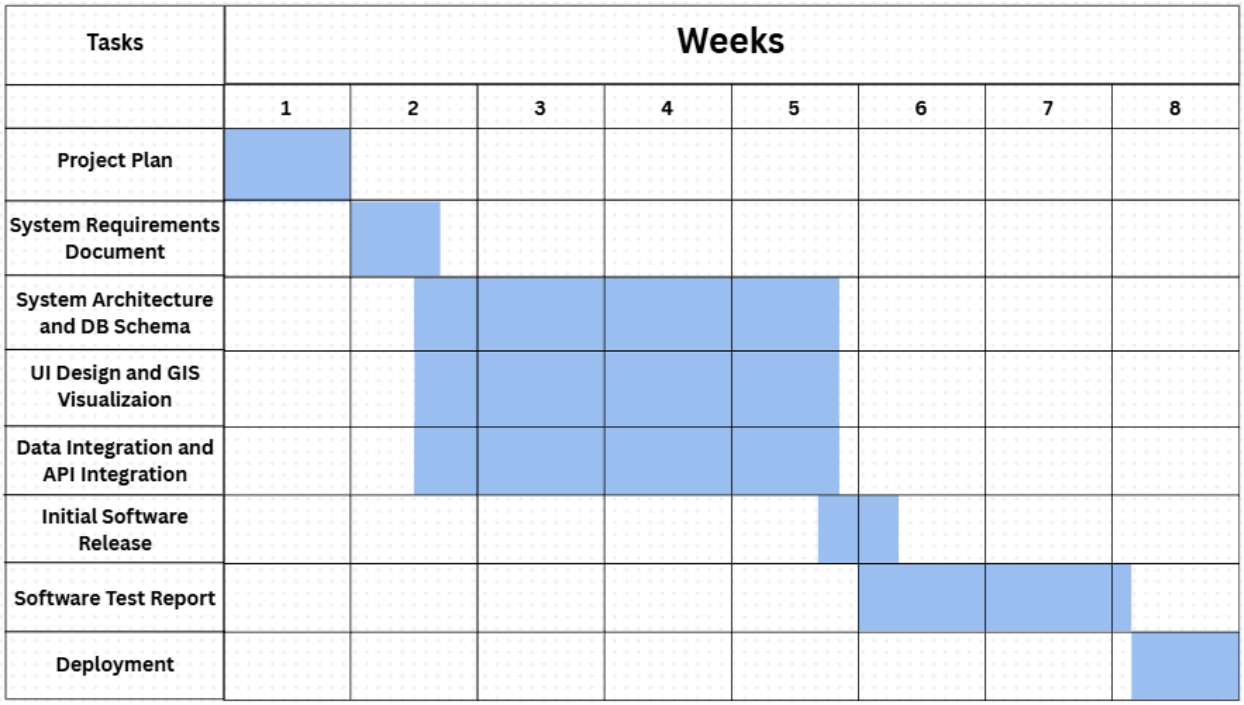


Figure 4: GIS Project Timeline

3.3 Project Metrics

To ensure that project progress, software quality, and performance remain on track and controlled throughout development, the following metrics will be monitored:

- **Schedule Variance:** The difference between actual project progress and the scheduled timeline at any milestone.
- **Number of Dynamic Web Pages:** Tracks the interactive interfaces that require user input or data processing, such as the map viewer.
- **Number of Persistent Data Objects:** Counts spatial database schemas, indexes, and location schemas being managed.
- **Number of External Systems Interfaced:** Measures data pipelines connecting to the main platform database.

3.4. Effort Breakdown

Team Member	Modules	Responsibilities
Mansur Khalil B. Batistil	System Architecture and DB Schema	Backend; design system architecture, prepare suitable database system and define appropriate database tables and relationships for the data, select and create appropriate web frameworks for use in creating the webApp, plan and prepare layouts for APIs, UI and DB and their interactions prior to creating these.
Justin Christian F. Chuah	UI Design and GIS Map Visualization	Frontend; implement interactive map interface, connect map to backend API, display AFAs depending on filters as map pins, present AFA data with certain user interactions
Luis Antonio P. Sarmiento	Data Integration and API Integration	Frontend and Backend; create data pipelines to read and process records from the database, build API

		endpoints for filtering, integrate GIS module into website

4. Project Risks

To lay out the potential risks that may affect the project, a risk table is used which lays out risks alongside its category, probability, impact, and a risk mitigation, monitoring, and management plan that can be carried out in order to deal with each risk. The category of the risks are divided into the following: PS - Project Size; SC - Stakeholder Characteristics; PD - Process Definition; DE - Development Environment; TE - Technology to be Built; ST - Staff Size and Experience. The impact of the risks are rated as follows: 1 - Catastrophic; 2 - Critical; 3 - Marginal.

Risks	**Category**	**Probability**	**Impact**	**RMMM**
Member inexperience	ST	70%	2	Monitor task progress
Insufficient time	PS	60%	2	Set milestones, review progress weekly
Requirements change while in development	PD	50%	2	Confirm requirements early, maintain constant communication with stakeholders for any new details
System integration issues	TE	50%	2	Test APIs early, verify compatibility with main website
Inadequate testing	PD	40%	2	Prepare test cases and conduct multiple tests before deployment
Scope change while in development	PS	40%	2	Set fixed project scope, assess time needed for additional features
Data integrity and migration issues	TE	40%	2	Validate and normalize Google Sheet records, ensure accurate results before migration
System performance issues	TE	40%	2	Optimize map loading and queries
Geographic data errors	TE	40%	2	Validate coordinates via

				testing
Integration of GIS to main website disrupts existing website features	TE	30%	1	Test integration in staging, have rollback option as backup
System maintenance and downtime	DE	30%	2	Monitor hosting and prepare system recovery procedures
Client resists system	SC	30%	3	Involve stakeholders in system demonstrations and collect feedback

References:

- Pressman, R. S., & Maxim, B. R. (2020). Software engineering: A practitioner's approach (9th ed.). McGraw-Hill Education.